

KEMSA Supply Chain Intelligence Platform

From reactive stockouts to a system that sees problems coming

Board Strategy Pitch | Michael Munga | Week 11

The Problem

- Facilities run out of essential medicines before anyone is warned — stockouts are discovered, not predicted.
- County payment risk is invisible until a county is already deep in debt.
- Surplus stock sits in one facility while another runs dry, because redistribution decisions are still manual and slow.

The Solution — Three Modules

- **Financial Risk Scorecard**

Weighted scoring (days overdue, debt amount, payment history) ranks every county's risk automatically, with a plain-language reason.

- **Stockout Forecasting**

30/60/90-day depletion forecasts per facility-medicine pair, flagged Critical / Warning / Watch / Healthy.

- **Redistribution Matching**

Finds the nearest facility with surplus stock using real geographic distance, and proposes a transfer for human approval.

How It Works

- Backend: FastAPI services over the existing analytics database — real data, not mock data.
- Frontend: Next.js dashboards, role- and scope-enforced — a Kiambu user cannot see Mombasa's data.
- Four views: National, County, Facility, and an executive Control Plane — each built for its own audience.
- Every redistribution suggestion requires a human Approve / Reject — automation assists, it does not replace the decision-maker.

Proof — Real Numbers, Not Projections

- Bungoma flagged at 91.8/100 risk vs. Bomet at 5.7/100 — the model separates real high- and low-risk counties correctly.
- A live match found a Kiambu facility with surplus stock 8.0 km from a Nairobi facility in shortage — real distance, real transfer, approved end-to-end.
- One quarter of real 2024 historical data shows KES 49.2M in commodity value across ~85,000 recommended transfers that could have been proactively redistributed.

The Financial Case

- Demonstrated quarterly opportunity: KES 49.2M (real historical data)
- Year-1 conservative capture (single-region pilot): KES 19.7M
- Year-1 platform cost: KES 2.4M
- **Net Year-1 benefit: KES 17.3M — ROI $\approx$ 720% — Payback $\approx$ 1.5 months**

Risks & Mitigation

- **Adoption risk: staff may not trust automated recommendations.**

Mitigation — human Approve/Reject on every transfer; nothing moves on its own in Year 1.

- **Forecast accuracy risk: consumption data quality varies by facility.**

Mitigation — tiered flags instead of binary alerts, plus review before high-value transfers.

The Ask

Approve a one-county pilot, with a Go/No-Go review on national rollout after 90 days of live data.

Value is already proven in the historical data — the ask is permission to run it live, with a human in the loop at every step.

Thank You

Questions & Discussion